AI as a co-creator and a design material: Transforming the design process

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Recent advancements in Artificial Intelligence (AI) have created new opportunities for incorporating AI into creative activities. Consequently, AI has become an increasingly significant tool in the design process, changing traditional workflows. This paper explores AI's role as both a co-creator and a design material, focusing on its impact on the ideation and evaluation stages of the design process. Through a transdisciplinary literature review, this study reveals that AI enhances creativity by providing inspirational stimuli, and supports evaluation and decision-making. However, it also introduces complexities such as cognitive overload and dependency. This paper emphasizes the need for design education reform, and training students as Designer Arbiters and Integrators to effectively collaborate with AI in a rapidly evolving technological landscape.
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Keywords: artificial intelligence, design process, creativity, human-AI collaboration, design education

1 Introduction

Artificial intelligence (AI) has become a powerful tool across many fields, including industries, transportation, finance, and healthcare (Rashid, Mullah, & Zain, 2020). AI refers to computer systems that reproduce human cognition by learning from data accessed from various sources and statistical models to recognize patterns and make decisions (Hassani, Silva, Unger, TajMazinani, & Mac Feely, 2020). As AI has evolved in recent years, its technologies are also becoming increasingly important in the field of design. AI presents unique opportunities and challenges that have significant implications for how designers work and innovate (Flechtner & Stankowski, 2023; Liao, Hansen, & Chai, 2020).

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Although design encompasses various categories and fields, such as graphic design, industrial design, architectural design, and interaction design, it is characterized by its process — the design process (Aspelund, 2015). Design is a multifaceted process that involves understanding the problems and the

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needs, generating innovative ideas, and transforming these ideas into practical and attractive solutions (Kim, 2011). According to Lazar (2018), this process can be divided into four stages:

1. Problem-Defining Stage
2. Ideation Stage
3. Evaluation Stage
4. Final Outcome Stage

The problem-defining stage is the initial phase where designers focus on researching, framing the context, understanding the constraints, and defining the problem and the goals. Next is the ideation stage, where designers generate a wide range of ideas and potential solutions. The evaluation stage often overlaps with the ideation stage, as designers frequently switch back and forth in a process described as "mode-shifting" by Lazar (2018, p. 3). During this phase, designers assess the feasibility and effectiveness of the generated ideas. This involves creating prototypes, conducting user testing, gathering feedback, and refining the ideas through a cycle of testing and iteration to ensure they meet the goals and user needs (Gmeiner, Yang, Yao, Holstein, & Martelaro, 2023). The ideation and evaluation stages can together be regarded as the creation stage, where creativity is a crucial aspect (see Figure 1). Zaidel (2014) defined creativity as "the introduction of something innovatively new and positive for society that goes beyond the familiar and accepted" (p. 1). The final outcome stage is where the selected tested prototype is finalized and prepared for implementation or production (Lazar, 2018).

AI has begun to play a significant role in this design process, and this role is not merely a technological shift but deserves to be examined from the perspective of psychological, neural-based, and educational influences on designers (Flechtner & Stankowski, 2023; Liao, Hansen, & Chai, 2020). The psychological perspective may help us understand how designers think and make decisions, while neuroscience may reveal how brain functions are affected by the

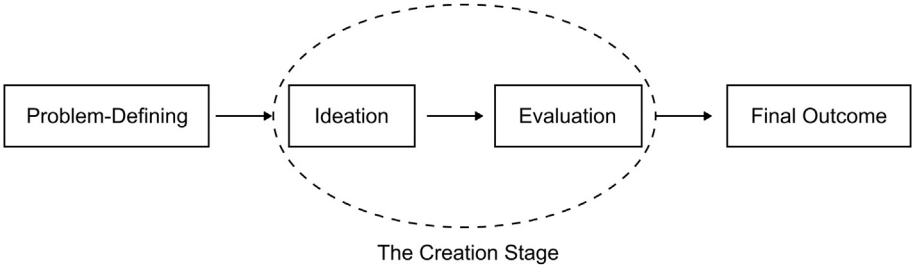


Figure 1 The design process

Note. *This figure was created by the author, based on Lazar (2018)*

inclusion of AI in the design process. The educational perspective can discover how educational strategies can enhance the collaboration between AI and designers. This transdisciplinary approach may provide a more nuanced understanding of the distinct cognitive stages involved in the design process, and is essential to understanding and optimizing the use of AI in design to ensure that designers can effectively collaborate with AI to enhance their work.

1.1 Background

The recent role of AI in the design process can be viewed from two main perspectives: AI as a co-creator and AI as a design material (Figoli, Mattioli, & Rampino, 2022). With the advancement of AI technologies, a new category known as Generative AI has emerged, which refers to "computational techniques that are capable of generating seemingly new, meaningful content such as text, images, or audio from training data" (Feuerriegel, Hartmann, Janiesch, & Zschech, 2023, p. 111). The ability of AI to generate meaningful content offers an opportunity for people to collaborate with AI in creative activities (Guo, Xiao, Wang, & Ji, 2023, pp. 9–13; Kim & Maher, 2023). For designers, Generative AI could act as a co-creator, providing numerous possibilities to inspire solutions, and bring changes to the design process (Chandrasekera, Hosseini, & Perera, 2024; Figoli, Mattioli, & Rampino, 2022; Guo, Xiao, Wang, & Ji, 2023, pp. 9–13; Kim & Maher, 2023; Liao, Hansen, & Chai, 2020).

On the other hand, AI has enabled otherwise impossible forms of interaction and a new dimension of personalization and adaptability, leading to the perspective of AI as a design material in design communities (Dove, Halskov, Forlizzi, & Zimmerman, 2017; Olsson & Väänänen, 2021; Windl, Feger, Zijlstra, Schmidt, & Wozniak, 2022; Yang, Steinfeld, Rosé, & Zimmerman, 2020). AI can be integrated into final products, creating AI-infused designs or systems that harness AI capabilities to directly influence the end users, such as "pattern recognition," "automated filtering," and personalized "search engines" (Amershi et al., 2019, pp. 1–2). The addition of AI to the design materials requires the designers to effectively envision new uses for AI and inevitably changes the conventional design process.

1.2 The problem

While there is substantial knowledge about the design process (e.g., Aspelund, 2015), the concept of creativity (e.g., Heilman, 2016; Zaidel, 2014), and AI's capabilities (e.g., Feuerriegel, Hartmann, Janiesch, & Zschech, 2023; Hassani, Silva, Unger, TajMazinani, & Mac Feely, 2020; Rashid, Mullah, & Zain, 2020), there is a notable gap at their intersection. Specifically, less is known about how these AI advancements impact designers' cognitive and neurocognitive processes. A few studies have separately examined AI as a co-creator (Chandrasekera, Hosseini, & Perera, 2024; Figoli, Mattioli, &

Rampino, 2022; Guo, Xiao, Wang, & Ji, 2023, pp. 9–13; Kim & Maher, 2023; Liao, Hansen, & Chai, 2020) and AI as a design material (Dove, Halskov, Forlizzi, & Zimmerman, 2017; Olsson & Väänänen, 2021; Windl, Feger, Zijlstra, Schmidt, & Wozniak, 2022; Yang, Steinfeld, Rosé, & Zimmerman, 2020), but there is limited exploration of their influence on the design process from a cognitive and neural-based perspective.

More specifically, the impact of AI on the ideation and evaluation stages of the design process—the critical phases of creation where ideas are generated and refined—remains a particularly important focus. Given the ongoing research interest into how the human brain engages in the creative thought process (Beaty, Benedek, Silvia, & Schacter, 2016; Benedek, Jurisch, Koschutnig, Fink, & Beaty, 2020), it is crucial to understand how AI affects this creation stage and influences designers' cognitive functions, such as creativity, problem-solving, and cognitive flexibility. This knowledge is essential to optimize human-AI collaboration in the future of design.

Moreover, while the need to integrate AI into design education is widely recognized, current literature often lacks practical details on how to achieve this. There is a clear necessity for educational strategies that can equip designers with the skills and knowledge required to effectively use AI in their work, while also addressing the rapidly evolving nature of AI. This research aims to address these gaps by analyzing the cognitive and neural-based impacts of AI on designers during the creation stage of the design process and by informing more effective educational strategies based on the findings.

1.3 Research question

How and to what extent does AI transform the creation stage (ideation/evaluation) of the design process, from a psychological, neural-based, and educational perspective?

2 Literature review

In order to respond to the research question, this literature review is divided into two main categories: (a) Design Process, Cognition, and Neuroscience, (b) AI in the Design Process. Section (b) is further subdivided into AI as a Co-Creator and AI as a Design Material.

2.1 Literature genres

This literature review primarily used peer-reviewed journal articles, but also includes conference proceedings, meta-analysis, systematic reviews, dissertation, books, and chapters. The research considered the fields of psychology, neuroscience, health and education—a transdisciplinary approach.

2.2 Design process, cognition, and neuroscience

This section of the literature review examines the cognitive and neural aspects of the design process. It focuses on how cognitive functions and processes contribute to ideation and evaluation stages. It also explores the activation of brain regions in the design process.

2.2.1 Creativity, inspiration, and fixation

Creativity is fundamental to the design process. Zaidel (2014) argued that "artistic talent (inborn) and skill (can be taught) are enmeshed with creativity" (p. 5). She added that artistic talent ranges widely, and creativity interacts at each level of this continuum, but in a healthy brain creativity might function independently of talent. Creativity relies not only on inborn talent, but also on "intact knowledge and conceptual semantic systems, healthy neural connectivity, and normal levels of neurotransmitters such as dopamine" (Zaidel, 2014, p. 6). She also suggested that while some neural structures for creativity exist in animals, the neural basis for creativity is more pronounced in humans, indicating that humans possess a more advanced capacity for creativity.

Enhancing creativity can involve various strategies. Gero and Milovanovic (2023) discussed an interesting way of enhancing creativity using neurofeedback. In their study, they provided real-time feedback about brain activation during the design process, helping designers sustain cognitive activation. They found that "participants who received the neurocognitive feedback produced twice as many ideas in the same time period as the group who received no feedback" (p. 16).

Inspiration and overcoming design fixation are also crucial for creativity. Design fixation, as described by Crilly and Cardoso (2017), is a state where "someone engaged in a design task undertakes a restricted exploration of the design space due to an unconscious bias resulting from prior experiences, knowledge or assumptions" (p. 6). They suggested that understanding and addressing design fixation can have profound implications for education and tool development. Educational strategies could focus on teaching design and creativity methods that combat fixation, while design tools could be developed to support designers in recognizing and overcoming fixation. These tools could provide feedback to designers about their thought processes and offer diverse stimuli to encourage broader exploration of the design space (Crilly & Cardoso, 2017). A relevant example is Brian Eno and Peter Schmidt's Oblique Strategies deck, which provides prompts designed to disrupt conventional thinking patterns and overcome creative blocks (Roy & Warren, 2019).

Goucher-Lambert, Moss, & Cagan (2019) explored the concepts of inspirational stimuli and analogical reasoning. According to this study, "analogical reasoning is generally defined as the process by which information from a source is applied to a target through the connection of relationships or representations between the two (source and target)" (p. 2). Inspirational stimuli provided to designers can facilitate the retrieval of useful concepts from memory, enhancing concept generation. Near-field stimuli (analogy that comes from the same or related domain), in particular, have been shown to sustain a more productive level of ideation and promote abstract thought (Goucher-Lambert, Moss, & Cagan, 2019).

In summary, creativity in the design process involves a complex interplay of cognitive processes and strategies to overcome fixation and enhance inspiration. These elements collectively contribute to the generation of innovative ideas and solutions. Understanding these factors sets the stage for further examining the stages of the design process through a cognitive lens.

2.2.2 Stages of the design process through a cognitive lens

As described earlier, Lazar (2018) outlined four stages of the design process: problem-defining, prototype generation, evaluating outcomes, and final outcomes. While all stages are important, this paper focuses specifically on the middle two stages, because they are most relevant to the creation stage, which is central to the research question. Therefore, the review will concentrate on Lazar's (2018) insights on these two stages (Figure 2). During the prototype generation stage, the main cognitive strategy is divergent thinking, characterized by "thinking that moves away in divergent directions and makes associations with ideas that are seldom directly connected" (Lazar, 2018, p. 3). This process, often synonymous with creativity, uses analogical reasoning, as mentioned previously, to connect present problems with similar past problems, enhancing creativity by mapping deeper structural connections rather than surface characteristics. The evaluating outcome stage involves convergent thinking, which focuses on analytical thinking and refining ideas into final solutions (Lazar, 2018).

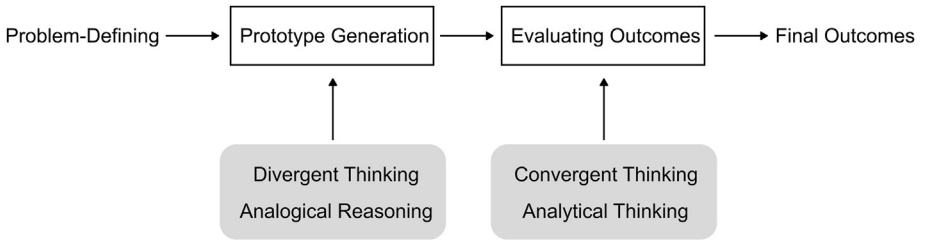


Figure 2 Cognitive processes in stages of the design process.

Note. *The design process highlighting the creative stages. This diagram is based on Lazar (2018)*

2.2.3 Neural basis of the design process

Lazar's (2018) research provided foundational insights into the neural processes involved in design, emphasizing the critical role of the brain's prefrontal cortex (PFC) during the generative phase of the design process. Khalil and Moustafa (2022) further contributed to this understanding by proposing a neural model that highlights the importance of the right PFC in creative tasks, particularly in idea generation and problem framing. The dorsolateral prefrontal cortex (DLPFC), as identified by Lazar (2018), is heavily involved in planning and correlates with areas responsible for mental imagery and self-generated representations. Lazar (2018) also suggested that the medial temporal lobe (MTL), including the hippocampus (Khalil & Moustafa, 2022), is also activated during idea generation, indicating its role in mental imagery, analogical processing, and generating novel ideas. During evaluation phases, the prefrontal areas and the default network—which integrates sensory input with visceral, autonomic, and hedonic information—are activated, supporting both analytical assessment and emotional evaluation (the "gut feeling") in the design process (Lazar, 2018, p. 3).

2.2.4 Team dynamics in design

Effective design thinking also requires supporting communication and collaboration within design teams. Gero and Milovanovic (2023, pp. 7–24) noted that "domain knowledge influences one's mode of thinking and engaging in project development", which can lead to "cognitive misalignments" (p. 18) between team members from different disciplines. Addressing these misalignments through better understanding and support can enhance collaboration in design teams (Gero & Milovanovic, 2023).

Overall, this section has reviewed the fundamental role of creativity in the design process, the strategies to enhance it, and the challenges posed by design fixation. It has also examined the stages of the design process, focusing on prototype generation and evaluation, and discussed the neural basis of design thinking and the dynamics within design teams. These insights into the cognitive and neural-based aspects of design set the foundation for examining how AI might transform the design process. The following section will review the role of AI in the design process.

2.3 AI in the design process

This section explores the role of AI in the design process, focusing on its dual functions as a co-creator and as a design material. The literature review again particularly concentrates on the creation phase in the design process, which includes the ideation and evaluation stages.

2.3.1 AI as a co-creator

2.3.1.1 Creativity enhancement through AI. Figoli, Mattioli, and Rampino (2022) have identified a future trend where designers increasingly collaborate with more-than-human intelligence, which refers to the advanced capabilities of AI that go beyond human cognitive limitations, such as processing vast amounts of data, identifying patterns at fast speed, and multi-tasking (Korteling, van de Boer-Visschedijk, Blankendaal, Boonekamp, & Eikelboom, 2021). Figoli, Mattioli, and Rampino (2022) argued that AI offers numerous potentialities during the design process, making it "an intriguing future collaborator" (p. 19). Integrating AI into the creative process in design aims to overcome human limitations and enhance capabilities, optimizing resource allocation and boosting creativity. They emphasized the significance of AI's relevance to design through the "random stimulus principle of lateral thinking" (Figoli, Mattioli, & Rampino, 2022, p. 31), which highlights the inclusion of unfamiliar conceptual elements to disrupt designers' preconceived notions and reasoning patterns. According to Figoli et al. this approach counters the risk of designers becoming fixated on a single paradigm, thereby stagnating in a limited solution space. By introducing external, sometimes random stimuli, AI helps designers reframe problems and discover innovative pathways.

Chandrasekera, Hosseini, and Perera (2024) further emphasized the role of AI in enhancing creativity in early design processes. They noted, "Generative AI algorithms, through machine learning and pattern recognition, can analyze extensive datasets of designs, enabling them to produce novel and often unorthodox design solutions" (Chandrasekera, Hosseini, & Perera, 2024, p. 3). They also discussed how AI-generated images can provide unexpected and inspiring visual cues, which are crucial for triggering lateral thinking—a cognitive process vital for innovative design. Lateral thinking is described as finding solutions in an indirect and creative way, utilizing non-obvious reasoning. The unpredictability of AI-generated images acts as a catalyst for this type of thinking (Chandrasekera, Hosseini, & Perera, 2024).

Kim and Maher (2023) supported Figoli, Mattioli, and Rampino (2022) and Chandrasekera, Hosseini, and Perera (2024)'s ideas through their experiment using the Collaborative Ideation Partner (CIP), "a co-creative design system that provides inspirational images based on conceptual similarity" (2023, p. 81). They found that AI-based inspirations, grounded in conceptual similarity, led to a slower decrease in the novelty, variety, and quantity of ideas over time compared to random inspirations. This finding validated their hypothesis that AI-based conceptually similar inspiration images enhance the number of new and novel ideas during design tasks, showing that AI can significantly boost creativity in design tasks (Kim & Maher, 2023).

Doshi and Hauser's study (2024) on writers investigated the impact of generative AI on individual creativity. In this study, writers were assigned a topic for a short story. For those receiving AI assistance, they could ask the AI to generate a three-sentence idea based on the assigned topic. The results showed that generative AI can significantly enhance individual creativity by providing new ideas and inspiration. However, it was also found that the diversity of novel content decreased, as the stories generated with AI assistance were more similar to each other compared to those created by humans alone.

2.3.1.2 Roles of AI in idea development. Figoli, Mattioli, and Rampino (2022) also discussed the integration of AI in the idea development phases of design, drawing on Liao, Hansen, and Chai (2020)'s framework. This framework outlines three potential roles for AI: "representation creation, empathy trigger, and engagement" (Liao, Hansen, & Chai, 2020, p. 512). In the role of representation creation, Liao, Hansen, and Chai (2020, as cited in Figoli, Mattioli, & Rampino, 2022) discussed how AI systems can "provide inspiration, widen design scope or trigger design actions by suggesting texts or images" (p. 34). The use of AI as a random trigger supports analogical thinking by transferring ideas across domains to generate innovative solutions (Liao, Hansen, & Chai, 2020). As an empathy trigger, Liao, Hansen, and Chai (2020) explained that AI can support the designer's descriptive thinking, valuable for building scenarios and expanding the scope of possible design ideas. AI does not replace the designer's sensitivity, particularly concerning emotional aspects, but instead provides new and unexpected information that can inspire the designer (Figoli, Mattioli, & Rampino, 2022). In terms of engagement, AI helps designers learn new insights through paraphrasing, modeling, and prototyping, aiding in the reframing of design problems. AI may detect when designers are stuck and provide instructions or questions to help overcome barriers. Self-answering and self-instruction triggered by AI can lead to reflection and development of design activities, preventing fixation (Liao, Hansen, & Chai, 2020).

Liao et al. further explored AI's potential in early design activities, including creativity and decision-making. They noted that existing design support systems often overlook critical stages of conceptual design, which involves high-order thinking, such as creative and reflective thinking. Their empirical observations suggested that AI-based inspirations can lead to a burst of new ideas, even when initial stimuli seem irrelevant, agreeing with the findings in the previous section (Liao, Hansen, & Chai, 2020).

2.3.1.3 Cognitive load in AI-enhanced design processes. Chandrasekera, Hosseini, and Perera (2024) conducted a study to examine the impact of AI on cognitive load and design creativity. Cognitive load is defined as "the relative demand imposed by a particular task, in terms of mental resources required" (American Psychological Association, n.d.). They used the NASA Task Load Index (NASA-TLX), "a multidimensional scale designed to obtain

workload estimates from operators based on six subscales: mental demand, physical demand, temporal demand, performance, effort, and frustration" (Chandrasekera, Hosseini, & Perera, 2024, p. 9). The study involved two groups: one using AI tools and the other working without AI assistance. The findings revealed that the AI-assisted group benefited from cognitive offloading as a result of using AI tools, enabling a more efficient allocation of their cognitive resources, rather than being slowed down by repetitive elements (Chandrasekera, Hosseini, & Perera, 2024).

2.3.1.4 Evaluation and decision-making. According to Schmidt, Elagroudy, Draxler, Kreuter, and Welsch (2024), AI can also enhance the evaluation processes in the design process. AI's ability to process and combine extensive data sets allows for rapid evaluation, weighing multiple factors in real-time. This supports complex decision-making processes by balancing various design variables dynamically (Schmidt, Elagroudy, Draxler, Kreuter, & Welsch, 2024). Additionally, AI facilitates heuristic evaluations by using general principles to assess usability, providing holistic feedback on system strengths, weaknesses, and opportunities for improvement. This helps designers gain a deeper understanding of user experience and iterate on their designs accordingly (Schmidt, Elagroudy, Draxler, Kreuter, & Welsch, 2024).

Jarrahi (2018) provided insights into the influence of AI in decision-making in an organizational setting. He highlighted the complementary strengths of AI and human decision-making. In ambiguous situations with no precedents, humans rely on intuitive decision-making, drawing from experience and personal judgment, which is hard to replicate by AI. However, AI has strength in handling complex situations requiring rapid processing of large information volumes. A collaborative approach, where AI tackles complexity with analytical methods and humans address uncertainty with creative and intuitive strategies, shows the vision of human-AI symbiosis. This approach views AI as a tool for augmenting human capabilities rather than replacing them, enhancing overall decision-making processes (Jarrahi, 2018).

2.3.1.5 Challenges in Human-AI collaboration. Figoli, Mattioli, and Rampino (2022) expressed a positive view of human-AI collaboration. They noted that AI's predisposition to perform simple and repetitive tasks more efficiently than humans can facilitate the design process, but requires proper task allocation, which involves assigning tasks based on the strengths of both AI and human designers, to ensure success. Figoli et al. also introduced the concept of "Designer Arbiter" (p. 74). This new role of the designer combines traditional design skills, intuition, and sensitivity with advanced critical analysis expertise, allowing designers to evaluate AI outputs and effectively implement them in the design process. Designers are freed from routine manual

activities, and their focus shifts to higher-level management, ensuring a harmonious and productive partnership (Figoli, Mattioli, & Rampino, 2022).

Yet, other studies have identified many challenges in human-AI collaboration (Gmeiner, Yang, Yao, Holstein, & Martelaro, 2023; Guo, Xiao, Wang, & Ji, 2023, pp. 9–13). A workshop by Guo, Xiao, Wang, and Ji (2023, pp. 9–13) explored the dynamics of co-creation between designers and AI. In traditional brainstorming teams, team members were able to participate proactively. However, in AI-assisted teams, AI-generated images suppressed the design team's engagement, which led to dissatisfaction with AI-generated results. Guo, Xiao, Wang, and Ji (2023, pp. 9–13) identified three modes of interaction between designers and AI: AI-inspired, visualizing using AI, and AI without augmented design. AI-inspired mode is where designers start with a preliminary idea and use AI-generated images for new insights, combining their ideas with AI inputs to define the concept. Visualizing using AI is where AI assists in visualizing a designer's matured concept, creating detailed representations that may exceed the designer's original vision. AI without augmented design is where AI generates detailed images from simple prompts early in the process, which can cause cognitive load on the designer and lead to reliance on AI outputs, losing the designer's own agency and independent thinking (Guo, Xiao, Wang, & Ji, 2023, pp. 9–13).

Gmeiner, Yang, Yao, Holstein, and Martelaro (2023) also addressed the challenges of co-creation with AI, in the context of engineering and industrial design. Their study revealed that effective co-creation with AI presents a significant learning curve for designers. Through a series of think-aloud studies with trained designers new to AI, they found that while AI tools could enable the creation and exploration of more complex designs in less time, most designers encountered substantial challenges in the process. The challenges were categorized into three themes: "Understanding and fixing AI outputs, working 'collaboratively' with the tools, and communicating their design goals to the AI" (Gmeiner, Yang, Yao, Holstein, & Martelaro, 2023, p. 8). Designers often found it difficult to share control with AI tools, feeling that the AI dominated the design process. As a result, they either accepted the poor results or stopped using the AI and did the work by hand (Gmeiner, Yang, Yao, Holstein, & Martelaro, 2023).

Recent research highlights the potential of mixed-initiative co-creativity (MI-CC) systems to address challenges in human-AI collaboration by fostering deeper, adaptive interactions. Unlike traditional co-creation tools, MI-CC systems enable iterative, bidirectional exchanges between human designers and AI, where both parties dynamically contribute to the creative process.

AI in the design process

Yannakakis, Liapis, and Alexopoulos (2014) demonstrated that such systems, like Sentient Sketchbook, act as active collaborators offering computational initiatives that guide designers toward novel solutions, disrupt cognitive fixation, and enhance lateral thinking. Expanding on this, Lin et al. (2023) emphasized the importance of a broader design space in MI-CC systems, which allows for diverse user-AI communication modes, including scrutability (probing the system) and explainability. These features preserve user agency by enabling designers to refine and guide AI outputs while adapting to individual expertise and preferences.

Overall, the exploration of AI as a co-creator in the design process highlighted its potential to enhance creativity, support lateral thinking, and manage cognitive load. However, human-AI collaboration presents challenges in maintaining human engagement, which require careful consideration of interaction modes, communication strategies, and adaptive tools. Switching perspectives, the next section will explore AI as a design material.

2.3.2 AI as a design material

2.3.2.1 Limited innovation in AI-infused design. Dove, Halskov, Forlizzi and Zimmerman (2017) noted that despite AI's potential, there has been a notable lack of innovation in its use as a design material. It has not experienced "a wealth of design innovation that other technologies have" (Dove, Halskov, Forlizzi, & Zimmerman, 2017, p. 278). Yang, Steinfeld, Rosé and Zimmerman (2020) indicated that it was necessary to examine from the perspective of how AI as a design material might hinder the design process.

2.3.2.1.1 Understanding AI capabilities. One of the fundamental challenges in utilizing AI as a design material is the difficulty designers face in understanding its capabilities. Dove, Halskov, Forlizzi and Zimmerman (2017) underscored the difficulties designers encounter in comprehending machine learning (ML), often describing it as "black magic" (p. 283). This was echoed by Flechtner and Stankowski (2023), who identified that the fundamental impediment in designing with AI is designers' lack of understanding of the technical capabilities of AI technologies. Similarly, Yang, Steinfeld, Rosé and Zimmerman (2020) highlighted that this lack of understanding impedes the initial stages of the design process, particularly brainstorming and sketching. Further, Windl, Feger, Zijlstra, Schmidt and Wozniak (2022)'s study indicated that a basic understanding of AI's abilities and limitations is essential for productive collaboration between designers and developers in AI projects.

Yang, Steinfeld, Rosé and Zimmerman (2020) categorized this type of challenge with the term "capability uncertainty" (p. 6). Capability uncertainty

refers to the ambiguities surrounding the functionalities of AI systems. Yang, Steinfeld, Rosé and Zimmerman (2020) noted that this uncertainty is particularly pronounced during the early design ideation stage when designers are attempting to understand potential AI applications. The absence of a comprehensive catalog of AI capabilities complicates this process, making it difficult for designers to assess the feasibility of their ideas, which hinders their creative process (Yang, Steinfeld, Rosé, & Zimmerman, 2020).

2.3.2.1.2 Lack of innovation due to reliance on existing functions. Yang, Steinfeld, Rosé and Zimmerman (2020) also suggested that due to a lack of comprehensive understanding of AI, designers "may choose to leverage existing AI libraries or pre-built models to address their design problem at hand" (p. 4). While these systems offer the advantage of freeing designers from data-related problems, they represent only a small subset of AI capabilities. Dove, Halskov, Forlizzi and Zimmerman (2017) also pointed out that designers struggle to understand AI technology and envision uses beyond existing applications. Designers using existing AI solutions to solve problems often encounter limited innovation potential, and this is the reason why design-led innovations in ML and AI are still rare (Flechtner & Stankowski, 2023). The cumulative difficulties in understanding, ideating, and innovating with AI present a clear challenge for designers (Dove, Halskov, Forlizzi, & Zimmerman, 2017).

2.3.2.2 Prototyping complexities. Yang, Steinfeld, Rosé and Zimmerman (2020) highlighted that while capability uncertainty poses challenges in understanding what AI can do, "Output Uncertainty" complicates how designers conceptualize the AI system's behaviors. Sketching and prototyping are keys to any design process, serving as crucial activities through which "designers understand what the technology is and can do, engage in creative thinking, and assess and improve on their designs" (Yang, Steinfeld, Rosé, & Zimmerman, 2020, p. 5). However, sketching and prototyping AI-infused products and systems is particularly difficult and cognitively demanding because prototypes need to reflect the complexity and uncertainty inherent in AI models (Windl, Feger, Zijlstra, Schmidt, & Wozniak, 2022). As a result, designers often choose to use "Wizard of Oz" methods, which involve a human operator simulating the behavior of the system being tested (Riek, 2012) for prototyping and testing (Windl, Feger, Zijlstra, Schmidt, & Wozniak, 2022). However, this method does not fully capture the technical constraints or the output complexity of actual AI systems (Flechtner & Stankowski, 2023). Hence, design processes driven by such methods risk producing "fictitious design possibilities rather than realistic products" (Flechtner & Stankowski, 2023, p. 73).

2.3.2.3 Modes of AI integration in the design process. When using AI as a design material in the design process, there are various approaches to integrating AI into the workflows. Windl, Feger, Zijlstra, Schmidt and Wozniak

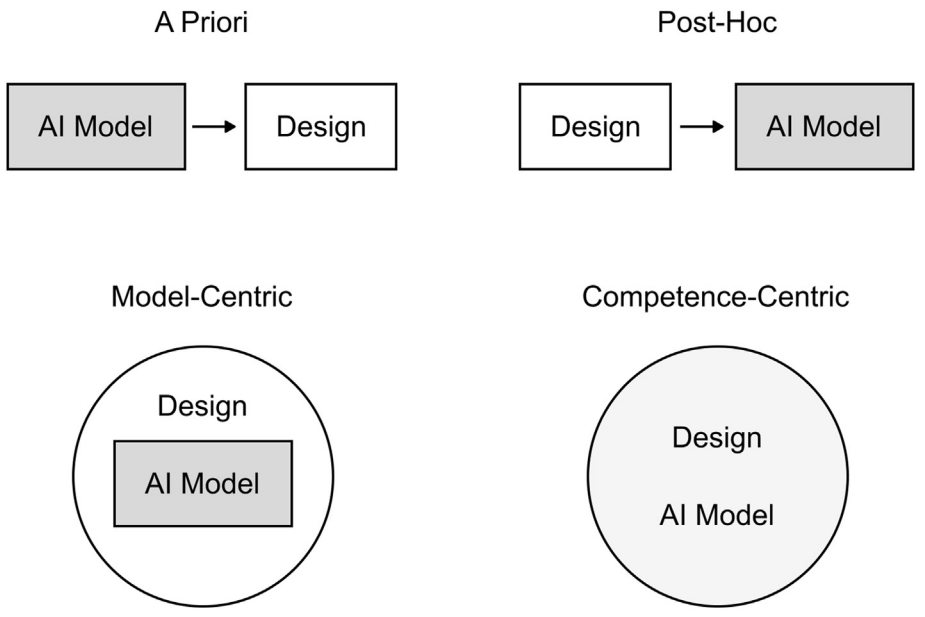


Figure 3 Four approaches to integrating AI in the design process.

Note. *This diagram is based on Windl, Feger, Zijlstra, Schmidt and Wozniak (2022)*

(2022) categorized these approaches into four primary models (see also Figure 3):

- A Priori: In this approach, the AI model is developed before the design begins, requiring designers to fulfill the model's requirements. This can lead to inflexibility.
- Post-Hoc: In this approach, the design is completed first, and the AI model is subsequently built to fit the design requirements. While this maintains design independence, it may overlook the technical constraints of the AI model.
- Model-Centric: This approach places the AI model at the core of the project, with designers deeply involved in almost all technical aspects. It fosters close collaboration between designers and AI specialists, integrating their expertise throughout the process.
- Competence-Centric: This approach emphasizes the diverse expertise within the team, with designers and AI specialists working in parallel and synchronizing their efforts.

Flechtner and Stankowski (2023) expanded on these concepts by outlining three categories of designers' engagement with AI in the design process, each requiring different levels of technical knowledge:

- Conceptual Approaches: These speculative methods aim to envision possible futures, raise questions, and foster critical debates about the social, cultural, and ethical implications of AI technologies. They are valuable for discussing the desirability of various technological futures.
- Alternating Approaches: Drawing from Windl et al.'s models, this category involves a design and development process where design and technical aspects are mostly separated, as seen in the *a priori* and *post-hoc* approaches described by Windl et al.
- Collaborative Approaches: In this tightly integrated process, design and technical development are closely intertwined, "requiring close collaboration among designers, data scientists, and AI specialists" (Flechtner & Stankowski, 2023, p. 73). The *model-centric* approach by Windl et al. exemplified this, where the AI model is central to the project and developed by an integrated team. This method requires designers to develop a deep understanding of AI functionalities, and establish a common language between design and AI.

2.3.2.4 Educational gaps. There is a clear educational gap in preparing designers to incorporate AI into their work. Dove, Halskov, Forlizzi and Zimmerman (2017) pointed out that the current design education cannot prepare the next generation of design graduates to use ML and AI in their work. Flechtner and Stankowski (2023) identified a broad range of requirements for future designers to creatively engage with AI. They emphasized the importance of designers being able to make informed decisions about using AI technologies, creatively applying existing technologies, being aware of potential consequences, and consciously using speculative approaches instead of treating AI as a "wildcard" (Flechtner & Stankowski, 2023, p. 74). Flechtner and Stankowski (2023) stressed the need for a basic understanding of AI's possibilities and limits, which is complicated by the rapidly changing nature of these technologies. This situation places high demands on design educators, who must possess a good knowledge of AI technology and continuously keep up with technological advancements to provide useful guidance to students in design studios or projects (Flechtner & Stankowski, 2023).

In summary, this literature review examined the impact of AI on the design process from multiple perspectives, including psychological and neurocognitive viewpoints. Finally, grounding these considerations in the context of educational design highlights a meta-design approach that unites the discussion in practical applications. This approach presents real dilemmas to be addressed in design education. In the following section, a detailed methodology leading to this literature review will be described.

3 Methodology

This paper is based on a transdisciplinary literature review encompassing the fields of design, psychology, neuroscience, and education to gain a comprehensive understanding of the impact of AI on the field of design.

3.1 Search strategy

The primary databases used for the literature search included Google Scholar and Harvard HOLLIS. The search terms applied encompassed a broad range of keywords, including "design," "design process," "designer," "cognitive," "cognition," "AI," "artificial intelligence," "generative AI," "creativity," "ideation," "brain," "neuro," and "education." Boolean operators (AND/OR) were employed to create combinations that refined search results and targeted studies that addressed the research question. For instance:

- Combining design-centric terms with AI-related terms (e.g., "design process" AND "AI" AND "creativity")
- Cross-referencing cognitive and neuroscience perspectives (e.g., "cognition" OR "neuro" AND "AI")
- Exploring educational implications (e.g., "design education" AND "artificial intelligence").

3.2 Screening and selection criteria

The initial search was conducted to encompass a 12-year period from 2012 to 2024, selected to capture the most recent developments in AI alongside foundational literature on the design process. This timeframe reflects the accelerating pace of research in AI while including influential earlier studies on design cognition and methodology. The search process involved:

1. Initial Filtering: An initial pool of studies was gathered by screening titles and abstracts for relevance to AI's integration into the design process. Articles that were clearly unrelated to the fields of interest or lacked substantial academic rigor were excluded.
2. Criteria for Inclusion: Studies were prioritized based on two primary criteria: (a) their relevance to the research question, particularly those discussing AI's role in ideation, evaluation, or its cognitive and educational implications, and (b) their recency, with emphasis on works published in recent years to reflect advancements in AI.
3. Snowballing Approach: References cited in initially identified articles were further explored to uncover additional relevant literature. This iterative process allowed for the identification of studies outside the initial search scope, enhancing the breadth and depth of the review.

3.3 Refinement and synthesis

The iterative review process resulted in a carefully curated body of literature. Although exact metrics were not maintained during this process, the final set of sources reflects diverse perspectives, including empirical studies, theoretical frameworks, and meta-analyses, offering a comprehensive view of the subject.

3.4 Limitations of the methodology

This review is not a systematic literature review but aims to provide a structured and reflective synthesis of existing research. The lack of specific numerical metrics and reliance on available literature may present limitations in comprehensively representing the field. Additionally, as AI technologies rapidly evolve, future studies may necessitate revisiting these findings to incorporate novel insights.

With the reviewed literature generated from the described methodology, the next step is to synthesize and analyze the findings. The following analysis will explore how AI impacts the creation stage of the design process, focusing on psychological, neural-based, and educational perspectives.

4 Analysis

This section synthesizes the findings from the literature review to explore the diverse ways in which AI influences the design process. The analysis examines AI's capacity to enhance creativity, improve decision-making, and manage cognitive load while also addressing the complex and often paradoxical aspects of AI integration. By offering a detailed examination of these dynamics, this section provides a deeper understanding of AI's transformative role in the design landscape.

4.1 AI supporting designers

The primary finding is that AI, when used as a co-creator, significantly enhances human creativity in the design process. This enhancement happens through several mechanisms, including overcoming fixation (Figoli, Mattioli, & Rampino, 2022; Liao, Hansen, & Chai, 2020), supporting lateral thinking (Chandrasekera, Hosseini, & Perera, 2024; Figoli, Mattioli, & Rampino, 2022; Kim & Maher, 2023), and managing cognitive load (Chandrasekera, Hosseini, & Perera, 2024; Liao, Hansen, & Chai, 2020), which collectively transform the creation stage of the design process. Firstly, we discover that AI promotes lateral thinking by providing unconventional solutions and perspectives, enabling designers to approach problems from innovative angles and to think outside the box (Chandrasekera, Hosseini, & Perera, 2024; Chandrasekera, Hosseini, & Perera, 2024; Kim & Maher, 2023). Two more thinking processes are associated with lateral thinking; analogical thinking and divergent thinking, and both of them are enhanced by AI. AI

stimulates analogous thinking by helping generate analogies and comparisons, which allows designers to discover new connections and insights that might not be immediately obvious (Liao, Hansen, & Chai, 2020). This process boosts creative exploration and helps in developing innovative solutions in the design process. In addition, not only do non-obvious linkages, but also near-field analogies provided by the AI that are semantically close to the designer's current context or ideas, help generate new ideas, thereby sustaining creativity during the design process, as proven in Kim and Maher's study (2023). Divergent thinking, the ability to diverge thoughts and generate multiple ideas about a topic in a short period, is also expected to be significantly enhanced by AI (Liao, Hansen, & Chai, 2020). AI's ability to quickly produce a wide range of creative inputs gives designers many possibilities to explore.

These thinking processes are also tied to overcoming fixation, which occurs when individuals become stuck in familiar patterns of thinking, and when designers' thinking is restricted due to unconscious biases from prior experiences. Liao et al.'s framework and empirical observations (2020) highlighted AI's potential to break cognitive rigidity and foster more dynamic ideation. This suggests that by introducing external stimuli and novel concepts, AI disrupts these conventional thinking patterns and promotes cognitive flexibility.

AI's potential to manage cognitive load during the design process is another critical finding. For instance, Chandrasekera et al.'s study (2024) has shown that AI tools can reduce the mental burden on designers by handling repetitive elements, allowing them to focus on more creative aspects of the design process. This cognitive offloading thus enables a more efficient allocation of designers' cognitive resources, which might enhance the overall creativity and productivity in the design process.

Moreover, the findings from the literature review imply that while AI greatly enhances the creative process, it does not replace human creativity but instead stimulates it. This suggests that AI does not remove designers from the design process but rather enhances their role. For example, Chandrasekera et al. (2024) suggested that AI's creativity is inherently different from human creativity due to the absence of consciousness, purpose, and the ability to put its creations in a meaningful context of human culture and history. Doshi and Hauser (2024) also proposed that AI may introduce a homogenizing effect and limit the range of unique ideas generated if AI is engaged too often on a collective basis. However, the inspiration provided by AI in writing (as examined in Doshi and Hauser's study) versus images (as used in the design context) might have different effects, as words often convey more specific ideas than images. Further research is needed to determine whether AI-generated image inspirations also lead to a homogenizing effect. In general, we can infer that AI is best considered a co-creator in the design process, working alongside human designers rather than replacing them.

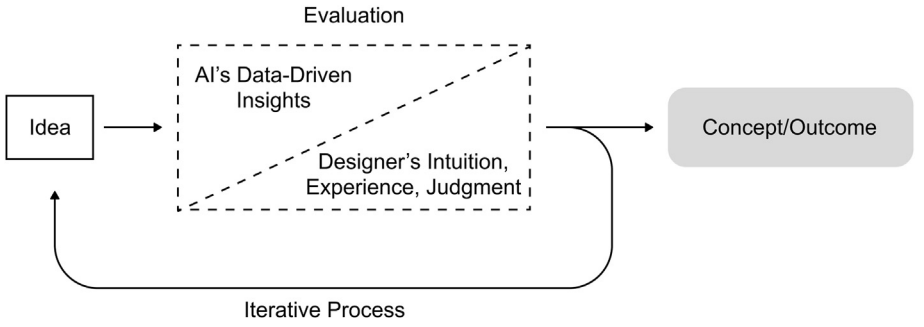


Figure 4 How AI transforms the evaluation and decision-making process.

Note. *This figure was created by the author*

Beyond its influence on ideation, AI also transforms the evaluation and decision-making process within the design workflow (Figure 4). AI enhances this process by providing data-driven insights and an objective analysis that leads to more informed decision-making. For example, AI algorithms can analyze large amounts of data to identify patterns and trends that might not be immediately obvious to human designers (Schmidt, Elagroudy, Draxler, Kreuter, & Welsch, 2024). This capability allows for more thorough evaluations of design concepts and could enhance designers' convergent thinking processes. Furthermore, this efficiency may enable designers to iterate more rapidly, testing and refining their ideas based on real-time feedback, thus leading to shorter creation cycles and faster innovation. Jarrahi (2018) highlighted that the introduction of AI in the design process encourages designers to use more intuition, experience, and personal judgment. This reflects that while AI focuses on data-intensive tasks, designers can focus on higher-order cognitive abilities such as emotional intelligence and contextual understanding. In summary, we may derive that AI's role in transforming the evaluation and decision-making stages of design not only accelerates the workflow but also allows designers to engage more deeply with the creative and intuitive aspects of their work.

4.2 Challenges in AI integration

While AI supports creativity, its integration into the design process presents challenges that complicate its role. The first challenge is the paradox that while AI may enhance creativity as a co-creator, it also poses risks of cognitive overload and dependency. This reveals a complex relationship between AI and human cognition. Firstly, while AI tools can handle repetitive tasks and manage some aspects of the design process, they can also introduce an overwhelming amount of information and options, as shown in Guo et al.'s study (2023). This might result in a cognitive overload as the designers find it hard to process and integrate immense information. Moreover, there is a risk of designers becoming overly dependent on AI, which can lead to a loss of creativity and

critical thinking skills. In the mode of interaction "AI without augmented design" described by Guo et al. (2023, p. 14), it was identified that as designers increasingly depend on AI, they may experience a loss of agency and feel less in control of the creative process. This loss of agency may reduce intrinsic motivation and lower the likelihood of innovative thinking, as designers could become accustomed to relying on AI instead of their own creative problem-solving skills. Gmeiner et al.'s research (2023) also supports the possibility that designers might develop a passive mindset, either by accepting poor results or abandoning the use of AI technology completely. Therefore, we can deduce that it is challenging to balance AI assistance with maintaining the designers' active engagement in the creation stage of the design process.

A second challenge is that when AI becomes a design material, rather than a co-creator, the innovation and creativity within the design process become much more limited in that designers often face challenges in understanding and using its full potential. According to Dove et al. (2017), many designers perceive AI as "black magic" and struggle to comprehend its capabilities and limitations (p. 283). This lack of understanding can severely hinder the ideation and innovation process because the designers may feel constrained by the technology rather than inspired by it. This cognitive load—where designers have to manage the complexities and uncertainties of AI capabilities—can overwhelm them leading to reduced cognitive flexibility. Moreover, Windl et al. (2022) pointed out that when designers prototype AI products, they often resort to the Wizard of Oz approach. However, this approach might fail to capture the full technical constraints and output complexity of real AI systems, and this will introduce bias in designers' understanding and expectations. This may result in potentially inaccurate mental models of how the AI product that they are designing will perform in real-world scenarios. Additionally, using AI as a design material tends to confine creativity within the boundaries of the AI's predefined capabilities due to a lack of complete understanding (Dove, Halskov, Forlizzi, & Zimmerman, 2017; Yang, Steinfeld, Rosé, & Zimmerman, 2020). This suggests that the cognitive flexibility of designers may be reduced and their ability to envision and create novel AI applications may become limited.

4.3 Underexplored areas in AI-design research

A major gap in current research is the limited existing research on AI's influence from a neuroscience perspective. While there is extensive research on the neural basis associated with design thinking and the design process (e.g., Beaty, Benedek, Silvia, & Schacter, 2016; Lazar, 2018; Zaidel, 2014), the understanding of how these neural processes are impacted when AI tools are incorporated into the design process is limited. The existing literature highlights the cognitive benefits and costs of AI, but there is a lack of empirical studies through brain imaging technologies that examine the neural mechanisms behind these benefits and possible neural costs.

Another underexplored area relates to design education, which must evolve to meet the challenges posed by AI integration in the design process. Through the findings from several studies (e.g., Dove, Halskov, Forlizzi, & Zimmerman, 2017; Flechtner and Stankowski, 2023; Gmeiner, Yang, Yao, Holstein, & Martelaro, 2023; Windl, Feger, Zijlstra, Schmidt, & Wozniak, 2022; Yang, Steinfeld, Rosé, & Zimmerman, 2020), it can be inferred that design education should evolve to help students understand how to effectively use AI and recognize its capabilities and limitations. Moreover, as Flechtner and Stankowski (2023) have pointed out, AI is rapidly changing. This suggests that it is also important to cultivate a growth mindset in students so they have resilience and adaptability in the face of the challenges of working with rapidly changing AI.

Design education should also aim at helping students find the optimal mode of interacting with AI, whether as a co-creator or as a design material (see Figure 5). In cases where AI is working as a co-creator, education should empower students to become "Designer Arbiters" (Figoli, Mattioli, & Rampino, 2022, p. 74) who can use AI to advance their design processes, rather than solely rely on AI. Guo et al. (2023, pp. 9–13) also advocated for the mode of interaction of AI-inspired or visualizing using AI. This means that AI provides inspiration and visualization but the designer retains control and creativity. When AI is used as a design material, Flechtner and

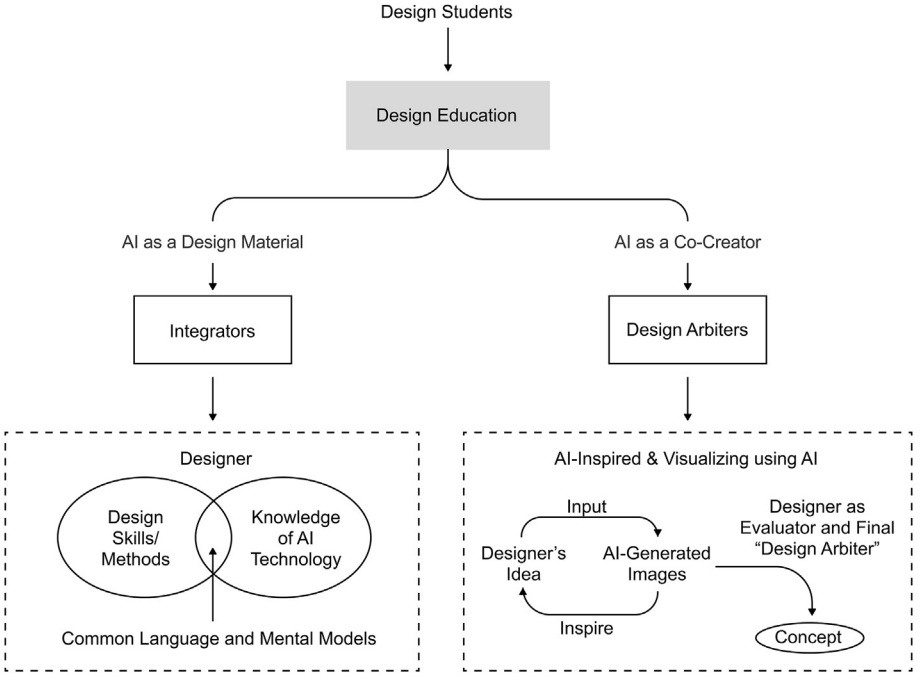


Figure 5 How design education should prepare students for AI integration.

Note. *This figure was created by the author*

AI in the design process

Stankowski (2023) highlighted the need to establish a common language and shared mental models between designers and AI technologies. This suggests that design students should be trained as integrators, developing a deeper understanding of technical knowledge in addition to design skills and methods.

In summary, the analysis of findings from the literature review reveals the complex impact of AI on the design process, showing the complexity in both the ways of integrating AI and the corresponding positive and negative implications. It offers important insights into how to answer the research question of this paper. The following conclusion will address the research question based on the analysis and discuss the implications and future research directions.

5 Conclusion

5.1 Answer to the research question

This paper explored the research question, "How and to what extent does AI transform the creation stage (ideation/evaluation) of the design process, from a psychological, neural-based, and educational perspective?" The analysis reveals that AI is able to profoundly influence the creation stage in the role of both co-creator and design material. Nevertheless, its impact is mixed as AI can augment but also complicate creative workflows. When functioning as a co-creator, AI significantly enhances the ideation and evaluation phases by generating stimuli that help spark creativity and help designers overcome cognitive fixation (Chandrasekera et al., 2024; Figoli et al., 2022; Liao et al., 2020). It fosters both analogical and divergent thinking (Kim & Maher, 2023; Liao et al., 2020), allowing designers to explore a broader spectrum of ideas and innovative solutions. AI also enhances evaluations by providing data-driven insights (Jarrahi, 2018; Schmidt et al., 2024) facilitating a collaborative design process where human designers and AI systems work together, combining human intuition with AI's computational power. However, these benefits come with challenges, such as cognitive overload and dependency on AI (Gmeiner et al., 2023; Guo et al., 2023, pp. 9–13), which may cause designers to have a loss of agency, reduced intrinsic motivation, and lowered likelihood of innovative thinking. On the other hand, when AI is utilized as a design material, the influence is currently found to be more negative. AI changes the design process in a way that requires the designer to have a deeper understanding of AI technologies (Dove et al., 2017; Flechtner & Stankowski, 2023; Windl et al., 2022; Yang et al., 2020). This necessity for designers to understand and manage AI's complex capabilities can introduce cognitive load, which can constrain creative expression if designers become overly focused on understanding the technology rather than exploring innovative ideas.

The potential impact of AI on the design process is considerable, but the exact extent of this impact is not yet fully understood. A more concrete and quantitative understanding requires further empirical studies. These studies should focus on measurable effects or results from the integration of AI in the design process. Such outcomes may help provide a clearer picture of the extent to which AI impacts the design.

5.2 Limitations of the study and recommendations for future studies

The first limitation of this study arises from the relatively young stage of research on AI and its integration into the design process. The dynamics between AI and design are still not fully understood or extensively researched. Consequently, this paper, based on literature reviews, is constrained by the availability and quality of prior research. Moreover, AI is a rapidly evolving technology, and the findings presented here may quickly become outdated as new AI advancements and applications emerge. Future research should continue to explore this area, and there may be a need for longitudinal studies on AI's impact on the design process.

The second limitation is the scarcity of studies relating to the neuroscience insights of AI in the field of design, particularly empirical studies using techniques like brain imaging scans. This gap limits the ability to fully understand the impact of AI from a neural-based perspective. Future studies are recommended to incorporate neuroimaging techniques to explore the brain activity associated with using AI tools in design tasks. Such research could provide insights into the neural mechanisms of human-AI collaboration.

A third limitation is that this paper has concentrated solely on the creation stage (ideation and evaluation stages) of the design process, due to time constraints. The paper did not consider the impact of AI integration on other stages, such as the problem-defining stage and the final outcome stage. Their exclusion limits the comprehensiveness of the study's findings. Future research should explore the role of AI across all stages of the design process to provide a more holistic understanding of AI integration in the design process.

Another limitation involves a lack of discussion on the potential biases introduced through the prompting mechanisms of generative AI. While AI has demonstrated the capacity to help designers overcome biases and cognitive fixation, as explored in this paper, the way designers craft AI prompts may inadvertently reintroduce these biases. For example, poorly constructed prompts or prompts influenced by designers' preconceptions may lead to outputs that mirror these biases or reinforce existing design patterns. Future research should investigate strategies for constructing prompts that balance context and creativity while minimizing the risk of introducing preconceptions.

Finally, this paper did not explore the ethical implications related to the use of AI in the design process. Future work should investigate potential biases introduced by AI tools and their impact on fairness and inclusivity. More specifically, when AI is used as a co-creator in the design process, the question of authorship needs to be considered. It is necessary to establish guidelines on how to attribute credit and ownership in collaborative AI-human design projects. When AI is used as a design material, research should address data privacy and protection, as AI systems often rely on large datasets that may contain sensitive or personal information.

5.3 General summary

This paper has explored the impact of AI on the design process, with a particular focus on its role as a co-creator and design material. The findings highlight the significant potential of AI to enhance creativity and support evaluation and decision-making within the design industry. However, these benefits come with challenges, including the risks of cognitive overload, dependency, and loss of agency. The paper emphasizes the importance of understanding these dynamics from psychological, neural-based, and educational perspectives to optimize AI's integration into design workflows.

The relevance of these findings goes beyond academic research, with practical implications for reform in design education. There is a need to develop curricula that equip future designers with the skills to collaborate effectively with AI, understanding its capabilities and limitations. This includes fostering a growth mindset and promoting cognitive flexibility to navigate the complexities of rapidly evolving AI technology and AI-enhanced design processes. Education should also enable the students to find the optimal mode of interacting with AI. Students need to be prepared to act as *designer arbiters* when interacting with AI as a co-creator, where they critically evaluate and integrate AI-generated inputs. In the second instance, when AI is used as a design material, students should be trained to become *integrators*, who effectively incorporate AI technologies into their designs while understanding and managing the technical constraints. By doing so, future designers will be better prepared to leverage AI in their design processes.

Author note

The author has no known conflict of interest to disclose.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Notes

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Data availability

No data was used for the research described in the article.

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